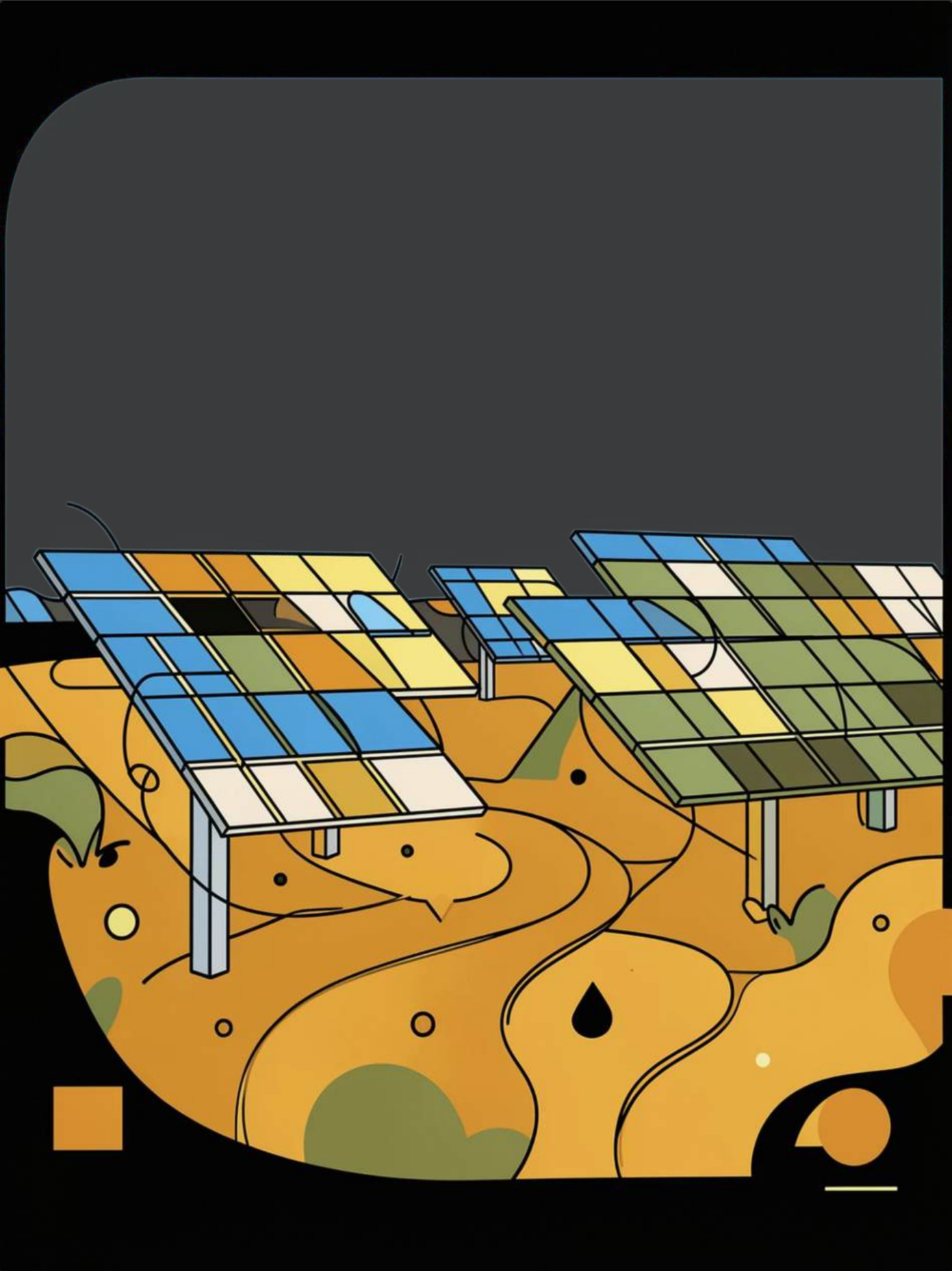


An illustration of solar panels on a desert landscape. The panels are tilted and arranged in rows, with some showing a grid pattern of blue, yellow, and green. The ground is a mix of orange and yellow, with some green bushes and a small black drop. The background is a dark grey gradient.

Solar Radiation & Power Prediction using XGBoost

Elg20 Project — A machine learning approach to forecasting solar energy output using real-world environmental datasets

GROUP
MEMBERS

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Overview

Solar radiation forecasting leverages environmental parameters to predict how much solar energy will be available at a given time and location. Accurate forecasts are essential for operating solar power systems efficiently and integrating them reliably into the electricity grid.

Efficient Generation

Optimise panel orientation and inverter settings based on predicted irradiance levels.

Energy Planning

Enable utilities and operators to schedule generation, storage, and demand response in advance.

Grid Stability

Reduce voltage and frequency fluctuations caused by intermittent solar output on the grid.

Key input parameters include **temperature**, **humidity**, **wind speed**, **atmospheric pressure**, and **cloud cover** — all of which directly influence the amount of solar radiation reaching the Earth's surface.



Project Objectives

01

Statistical Analysis

Compute descriptive statistics — minimum, maximum, and standard deviation — across all features for both datasets to understand data spread and variability.

02

Define Input–Output Relationships

Identify which environmental variables serve as predictors (inputs) and map them to the target variable — solar radiation in W/m^2 .

03

Apply XGBoost Algorithm

Train an XGBoost regression model on each dataset and generate solar radiation predictions using the optimised ensemble tree structure.

04

Compare Datasets

Evaluate and contrast model performance across two independent real-world datasets to assess generalisation and robustness of the approach.

Data Sources

Two independent, publicly available datasets were used to train and evaluate the XGBoost model. Both contain meteorological and solar parameters essential for energy prediction.

Dataset 1 — Figshare

Sourced from **Figshare**, an open-access scientific data repository curated for environmental and energy research. This dataset provides time-series measurements of meteorological parameters alongside recorded solar radiation values.

- High-quality, research-grade data
- 8,976 time-series records
- Solar Radiation (W/m^2) as output

Dataset 2 — Kaggle

Sourced from **Kaggle**, a widely used platform hosting real-world datasets and machine learning challenges. Chosen for its diversity and accessibility within the ML community.

- Community-validated dataset
- Rich meteorological feature set
- Compatible input-output structure

 Both datasets contain the same core meteorological features, enabling a direct side-by-side comparison of model performance under different real-world conditions.

Algorithm — XGBoost

XGBoost (Extreme Gradient Boosting) is a supervised machine learning algorithm built on an ensemble of decision trees. It iteratively trains new trees to correct the errors of previous ones, reducing prediction loss at each step — a process known as *gradient boosting*.

Why XGBoost for Solar Prediction?

→ Nonlinear Relationships

Captures complex interactions between temperature, humidity, cloud cover, and radiation that linear models cannot express.

→ High Accuracy

Consistently outperforms traditional regression and single-tree models on tabular datasets.

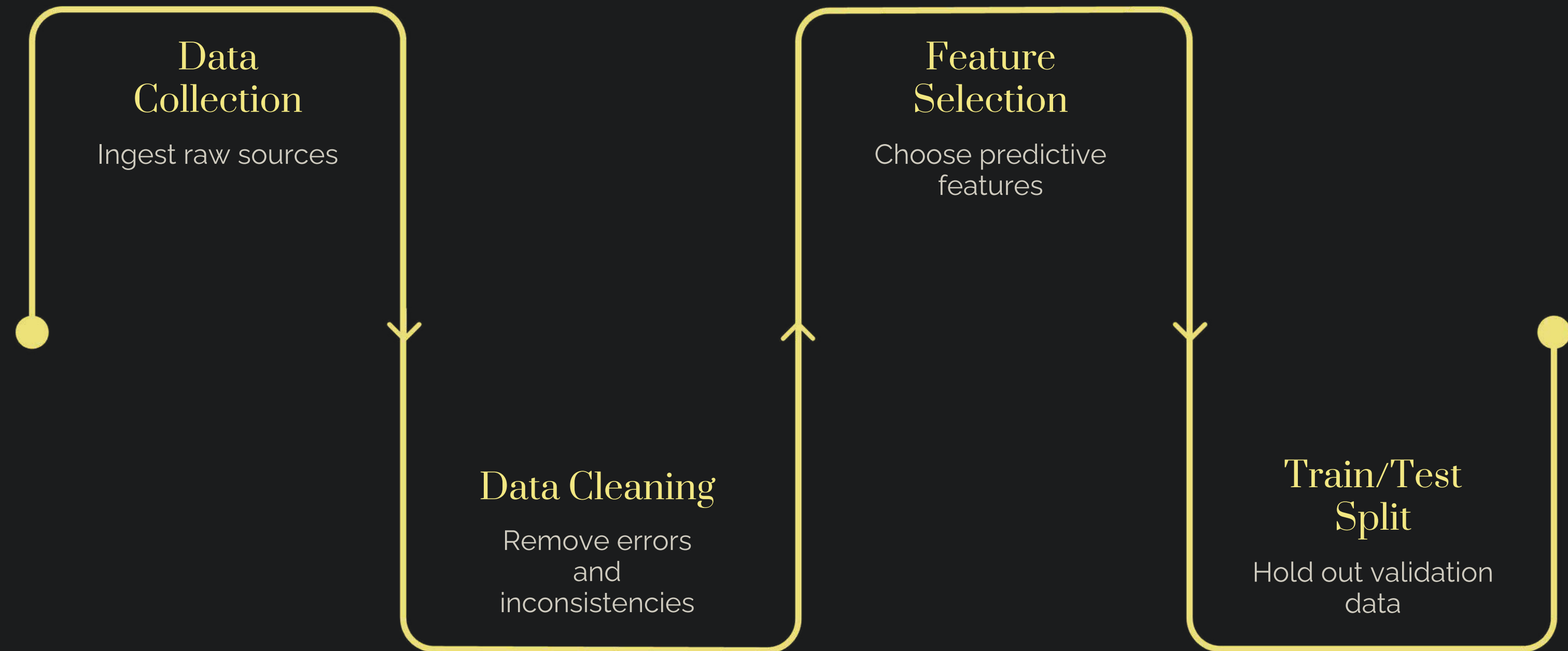
→ Scalability

Efficiently handles large datasets with thousands of records and multiple features without significant performance degradation.



Model Pipeline

The end-to-end workflow follows a structured machine learning pipeline — from raw data ingestion to final evaluation. Each stage is critical for ensuring the model learns meaningful patterns and generalises well to unseen data.



The pipeline ensures data integrity at every stage. Raw meteorological readings are cleaned to remove outliers and missing values, relevant features are selected based on correlation with solar radiation, and the model is evaluated using RMSE to quantify prediction accuracy against held-out test records.

Dataset 1 — Description

The Figshare dataset is a structured time-series collection of environmental sensor readings paired with simultaneously recorded solar radiation measurements, making it well-suited for supervised regression modelling.

Property	Value
Source	Figshare (Open-Access Scientific Repository)
Total Records	8,976
Data Type	Time-Series (Environmental Sensor Readings)
Target Variable	Solar Radiation (W/m²)
Input Features	Temperature, Humidity, Wind Speed, Pressure, Cloud Cover
Application	Supervised Regression using XGBoost

Input & Output Variables — Dataset 1

Five environmental parameters serve as model inputs, each capturing a different physical aspect of atmospheric conditions that collectively determine solar irradiance at ground level.



Temperature (°C)

Ambient air temperature influences atmospheric transparency and solar energy absorption.



Humidity (%)

Relative humidity affects cloud formation and water vapour absorption of solar radiation.



Wind Speed (m/s)

Wind disperses cloud cover and influences surface temperature, indirectly affecting irradiance.



Pressure (hPa)

Atmospheric pressure correlates with weather system patterns that modulate solar input.



Cloud Cover (%)

The most direct driver of solar attenuation — dense cloud cover sharply reduces W/m^2 at ground level.



Output: Solar Radiation (W/m^2) — the total shortwave irradiance reaching the surface, predicted by the XGBoost regression model.

Statistical Measures — Definitions

A comprehensive explanation of the three key statistical measures used in the analysis.

Minimum (Min)

Definition: The smallest value in a dataset

$$Min = \text{smallest value in } \{x_1, x_2, \dots, x_n\}$$

Example: Temperature range -13.9°C to 33.9°C, Min = -13.9°C

Interpretation: Represents the lower bound of the data range

Maximum (Max)

Definition: The largest value in a dataset

$$Max = \text{largest value in } \{x_1, x_2, \dots, x_n\}$$

Example: Temperature range -13.9°C to 33.9°C, Max = 33.9°C

Interpretation: Represents the upper bound of the data range

Standard Deviation (σ or SD)

Definition: Measures how spread out data points are from the mean

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$

Where: x_i = individual value, μ = mean, N = number of values

Example: Solar Radiation Std Dev = 221.93 W/m²

Interpretation: Higher SD = more variability; Lower SD = data clustered around mean



These three measures provide a complete picture of data distribution. Min and Max show the range, while Standard Deviation quantifies the spread around the mean, helping us understand data variability.

Statistical Analysis — Dataset 1

Descriptive statistics reveal substantial variability across all parameters, confirming the need for a powerful nonlinear model like XGBoost. Notably, solar radiation spans from 0 to 928 W/m² with a high standard deviation of 221.93, reflecting strong diurnal and seasonal patterns in the data.

Parameter	Min	Max	Std Dev
Temperature	6.98 °F	33.9 °C	8.52 °C
Humidity (%)	22.17	100	17.89
Wind Speed (m/s)	0	40	5.58
Pressure (hPa)	989	1048.5	8.18
Cloud Cover (%)	0	100	32.24
Solar Radiation (W/m²)	0	928	221.93

📌 The high standard deviation of solar radiation (221.93 W/m²) relative to its mean indicates that the target variable is highly non-uniform — a key justification for using ensemble tree methods over linear regression.

Model Results & Interpretation

Model Evaluation Metrics

To evaluate the performance of our XGBoost model, we used three important metrics:

RMSE (Root Mean Square Error)

Measures overall prediction error magnitude.

MAE (Mean Absolute Error)

Shows average absolute difference between actual and predicted values.

R² Score (Coefficient of Determination)

Indicates how well the model explains variance in data.

RMSE measures the average magnitude of prediction errors in the same units as solar radiation (W/m²). A lower RMSE indicates better model fit.

Key Takeaways

Moderate Prediction Accuracy

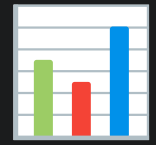
An RMSE of 134.52 W/m² reflects reasonable performance given the wide 0–928 W/m² range of the target variable, equivalent to roughly 14.5% of the full-scale range.

High Target Variability

The large standard deviation of solar radiation (221.93 W/m²) poses an inherent forecasting challenge. XGBoost manages this variability better than traditional linear models.

Next Steps

Hyperparameter tuning, feature engineering, and comparison with Dataset 2 results will further refine model performance and validate generalisability.



RMSE (Root Mean Square Error)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



Example:

Actual vs Predicted values:

- Actual: 100, Predicted: 90 → Error: 10

- Actual: 200, Predicted: 210 → Error: -10

Squared errors: (100, 100)

Mean: 100

RMSE = $\sqrt{100}$ = 10

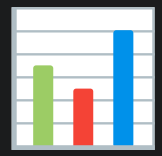


Our Result:

RMSE = 134.60



Indicates average prediction error magnitude (sensitive to large errors)



MAE (Mean Absolute Error)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Example:

Actual vs Predicted values:

- Actual: 100, Predicted: 90 → Error: $|10| = 10$
- Actual: 200, Predicted: 210 → Error: $|-10| = 10$

Absolute errors: (10, 10)

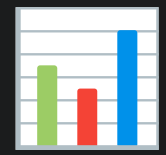
$$MAE = (10 + 10)/2 = 10$$



Our Result:

$$MAE = 87.44$$

 Shows average absolute error (easier to interpret than RMSE)



R² Score (Coefficient of Determination)

$$R^2 = 1 - \left(\frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \right)$$



Interpretation:

- R² = 1: Model predicts perfectly
- R² = 0.5: Model explains 50% of variance
- R² = 0: Model is no better than random
- R² < 0: Model performs worse than baseline

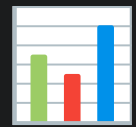


Our Result:

R² = 0.643



Model explains 64.3% of variation in data (good but not perfect fit)



Model Performance Summary



Key Results:

RMSE = 134.60

→ Moderate error magnitude (sensitive to large deviations)

MAE = 87.44

→ Average prediction deviation in W/m^2

R^2 Score = 0.643

→ Model explains 64.3% of variance in solar radiation



Interpretation:

The model demonstrates reasonable predictive capability given the high variability in solar radiation data (std dev: 221.93 W/m^2). The RMSE of 134.60 represents approximately 14.5% of the full measurement range (0-928 W/m^2), indicating moderate accuracy suitable for practical solar energy forecasting applications.



Strengths:

- Captures nonlinear relationships in environmental data
- Performs better than linear regression models
- Provides actionable predictions for energy management

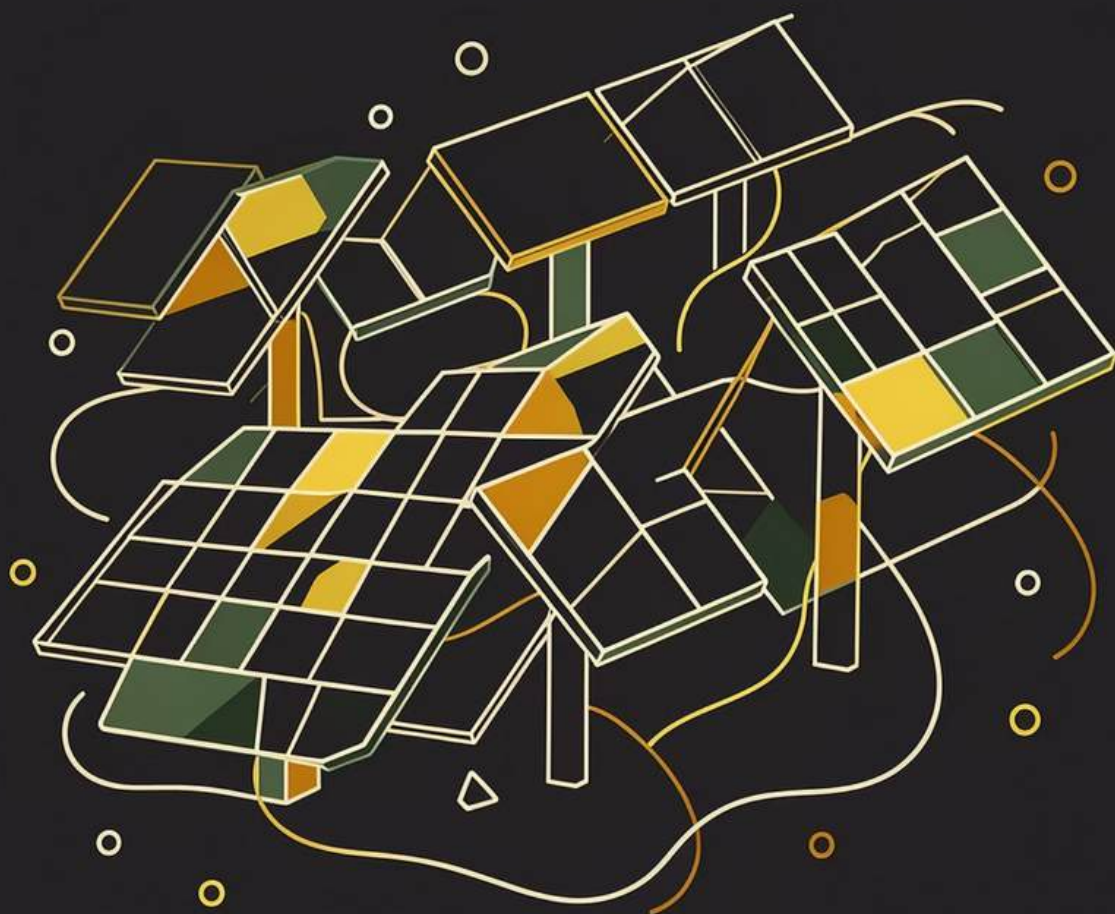


Limitations:

- Cannot explain ~36% of variance
- Sensitive to extreme weather conditions
- Requires continuous model refinement with new data

Dataset 2 — Description

The Kaggle dataset provides real-world solar power generation metrics alongside meteorological conditions, offering a distinct challenge for predicting actual power output rather than just irradiance.



Property	Value
Source	Kaggle (Community-Validated Dataset Platform)
Total Records	4,213
Data Type	Real-world solar power generation data
Target Variable	Generated Power (kW)
Input Features	Temperature, Humidity, Wind Speed, Pressure, Cloud Cover, Solar Radiation
Application	Supervised Regression using XGBoost

Input & Output Variables — Dataset 2

An explanation that Dataset 2 includes solar radiation as an input feature (unlike Dataset 1), making it a two-stage prediction problem.



Temperature (°C)

Ambient air temperature influences atmospheric transparency and solar energy absorption.



Humidity (%)

Relative humidity affects cloud formation and water vapour absorption of solar radiation.



Wind Speed (m/s)

Wind disperses cloud cover and influences surface temperature, indirectly affecting irradiance.



Pressure (hPa)

Atmospheric pressure correlates with weather system patterns that modulate solar input.



Cloud Cover (%)

The most direct driver of solar attenuation — dense cloud cover sharply reduces W/m^2 at ground level.



Output: Generated Power (kW) — the actual electrical power output from the solar installation, predicted by the XGBoost regression model using both weather and radiation inputs.

Key difference: Solar radiation is an input here, not the target. This reflects real-world power plants where radiation is measured and used to predict actual power generation.

Statistical Analysis — Dataset 2 (Environmental Parameters)

A paragraph explaining that environmental parameters in Dataset 2 show similar ranges to Dataset 1 but with some notable differences in variability.

Parameter	Min	Max	Std Dev
Temperature	22.37 °F	34.90 °C	8.85 °C
Humidity (%)	7	100	23.53
Wind Speed (m/s)	0	61.18	9.88
Pressure (hPa)	997.5	1046.8	7.02
Cloud Cover (%)	0	100	42.84

❏ Cloud cover shows notably higher variability (std dev 42.84%) in Dataset 2 compared to Dataset 1 (32.24%), suggesting more dynamic weather patterns. Solar radiation variability is also higher (278.46 vs 221.93), which directly contributes to the increased difficulty in predicting power output.

Statistical Analysis — Dataset 2 (Power Output)

Statistical analysis of Dataset 2 reveals the extreme variability inherent in real-world solar power generation. This variability reflects the complex interplay of meteorological conditions and their direct impact on actual power output.

Parameter	Min	Max	Std Dev
Generated Power (kW)	0.0006	3056.79	937.96

-  The extremely high standard deviation (937.96 kW) relative to the mean indicates that power generation is highly non-uniform across different times and weather conditions. This high variability poses a significant forecasting challenge and explains why Dataset 2 results in higher prediction errors than Dataset 1.

Model Results & Interpretation — Dataset 2

Model Performance – Dataset 2

795.96
RMSE
kW

662.73
MAE
kW

0.31
R² Score
Model explains 31% of data variation

Interpretation

RMSE and MAE are relatively high
Large errors indicate substantial deviation between predicted and actual power output.

Large variation in power prediction
The target variable is highly variable, which makes accurate forecasting more difficult.

R² = 0.31
The model explains 31% of the variation in the data.

Insight: Power generation is highly non-linear and variable. Environmental factors alone are not sufficient for perfect prediction.



Why is Error High?

Reasons for Higher Error

Power output depends on multiple factors:



Solar Panel Efficiency

The efficiency of converting solar radiation to electricity varies with temperature and panel condition



System Losses

Inverter losses, wiring losses, and other system inefficiencies reduce output

Data Characteristics:

- Dataset contains high variability (Std Dev $\approx$ 938 kW)
- Missing advanced influencing parameters in model
- Real-world complexity not fully captured by environmental sensors alone

Key Point:

"Prediction becomes harder when output depends on multiple hidden factors and system-specific characteristics"



Angle of Incidence

The angle at which sunlight hits the panels affects energy capture



External Environmental Noise

Weather patterns, dust, shadows, and atmospheric conditions introduce unpredictability

Comparison with Dataset 1

Dataset Comparison

Metric	Dataset 1 (Radiation)	Dataset 2 (Power)
RMSE	134.60 W/m²	795.96 kW
MAE	87.44 W/m²	662.73 kW
R² Score	0.643	0.31
Output Range	0–928 W/m²	0–3056.79 kW
Std Deviation	221.93 W/m²	937.96 kW
Error as % of Range	14.5%	26.0%
Prediction Complexity	Single-stage	Single-stage

Key Observations:

- 

Dataset 1 (Solar Radiation):

 - Better accuracy and model fit
 - Lower variability in target variable
 - Direct relationship with environmental inputs
 - More predictable patterns
- 

Dataset 2 (Power Generation):

 - More complex prediction problem
 - Higher output variability
 - Additional system-dependent factors
 - Requires more sophisticated modeling approaches

Conclusions & Limitations

✓ Key Conclusions

XGBoost Effectiveness

XGBoost successfully captures nonlinear relationships between environmental parameters and solar energy outputs across both datasets, outperforming traditional linear regression approaches.

Environmental Influence

Temperature, humidity, wind speed, pressure, and cloud cover collectively exert strong influence on both solar radiation and power generation, validating the multi-feature approach.

Real-World Complexity

Dataset 2 demonstrates that predicting actual power output is more challenging than predicting radiation alone, reflecting the practical complexities of solar installations.

⚠ Limitations & Future Work

High Variability Challenge

Extreme variability in power output (std dev 937.96 kW) limits prediction accuracy. Ensemble methods and hyperparameter tuning could improve performance.

Temporal Dependencies

The current model treats each record independently. Incorporating time-series features (lag variables, rolling averages) could capture temporal patterns.

Generalization

Model performance on unseen data from different geographic locations or seasons remains untested. Cross-validation across diverse conditions is recommended.



Thank You

Questions & Discussion

We appreciate your attention and welcome any questions about our solar radiation and power prediction analysis using XGBoost.

Manusha | Lokesh | Harshavardhini | Kailash | Sai Kumar

Elg20 Project